

D. Critical-difference diagram, F1 (minority) · N = 60 dataset×IR blocks · IR ∈ {5, 10, 20} · Friedman p = 4.0e-06

CD = 0.61

1 2 3 4



signature-guided, aligned step (1.92)

GPF-guided, aligned step (2.26)

SMOTE (raw) (2.88)

GPF-guided, plain step (2.95)